



Calculation Sheet

Calculation header

Identifier

*New calculation (1)***Medium selection and state**

Medium

Water

State

*Liquid***Operating data**☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	80.0	0.0	0.0	°C
p1	10.0	0.0	0.0	MPa(g)
○ p2	6.0			MPa(g)
● Δp	4.0	0.0	0.0	MPa
Kv	78.111			m³/h
○ qm	488,110.0			kg/h
● qv	500.0	0.0	0.0	m³/h
s/s100	99.389			%
LpAe	113.0			dB(A)
	Incipient cavitation			

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ρ1	976.22	0.0	0.0	kg/m³
ρ2	976.22			kg/m³
● η1	0.357	0.0	0.0	mPa s
○ ν1	0.3657			mm²/s
pv1	0.47414			bar(a)
pv2	0.48895			bar(a)
tv1	311.74	99.974	99.974	°C
cF1	1,574.1			m/s





Calculation Sheet

Pipe downstream of valve

Size class downstream of valve	NPS2	2"
Schedule downstream	SCH2	STD

Valve configuration

Valve type	Rotary plug valve
Trim type	Eccentric plug valve
Flow direction	FTO
Valve performance class	Heavy duty valve
Protection	Hardened seat/plug
Low-noise design	n.a.

Valve data

Basic characteristic		Equal percentage	
Nominal flow coefficient	Kvs	80.0	m ³ /h
☐ Lift stopper			
Selected valve size	NPS	2"	
Pressure class	class	class 600	
Maximum operating pressure	p,max	105.0	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL ²	0.714	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	0.603	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	0.283	-
Valve style modifier (Cv/Cv100 = 1)	Fd	0.44	-

Load-dependent values

	Maximum flow	Mean flow	Minimum flow	
s/s100	99.389			%
u2	68.525			m/s
LpAe	113.0			dB(A)
Ri	11.0			-

Noise calculation

Calculation standard (noise, liquid)	IEC 60534-8-4 (2015)
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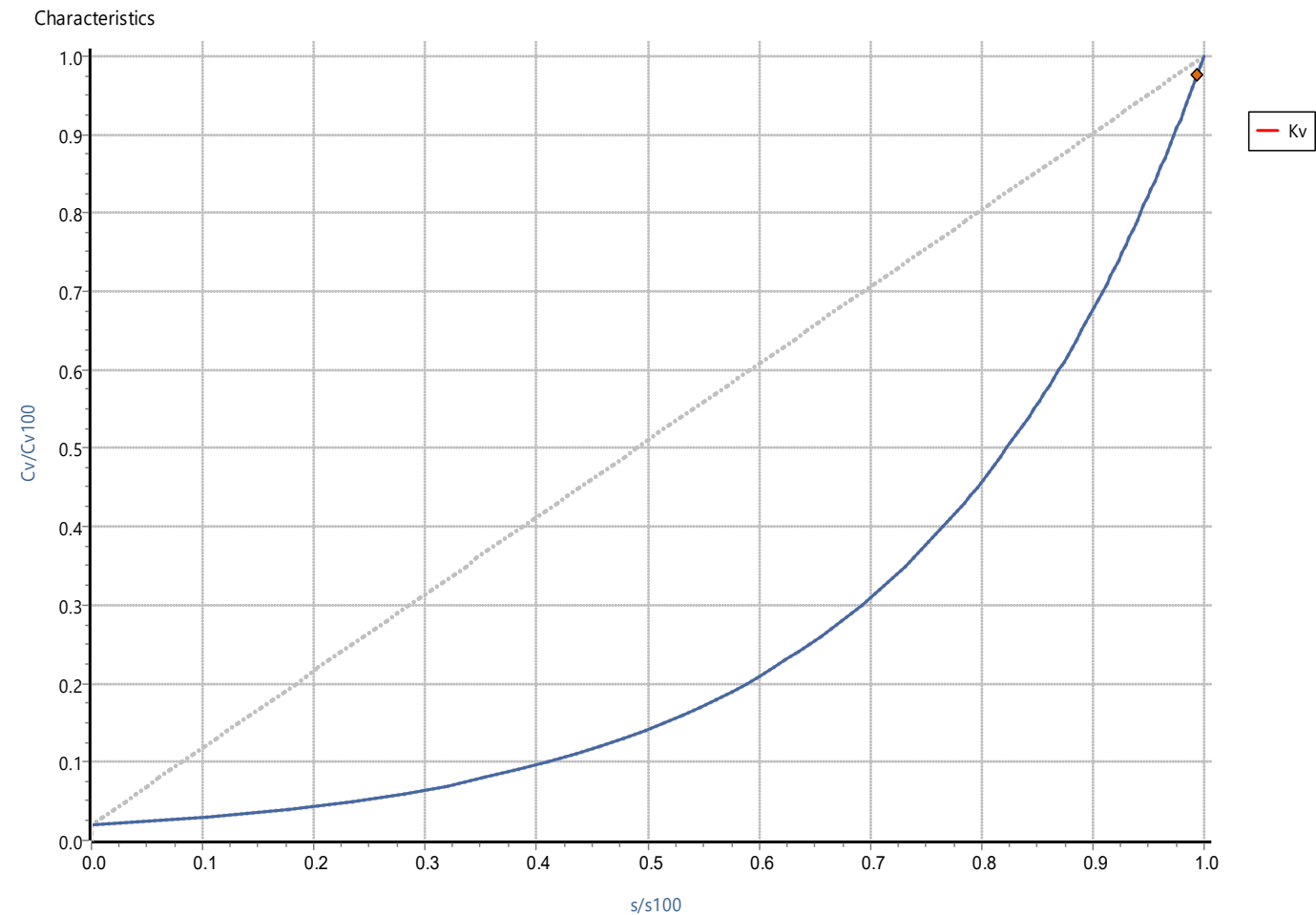




Calculation Sheet

Noise prediction data

Low-noise design		n.a.	
Sound velocity	cF1	1,574.1	m/s
Pressure level	LA1	182.0	dB
Pressure level	LA2	12.2	dB
Sound pressure level	Lpi	195.0	dB
Transmission loss	TL	-75.3	dB
Sound power level (A-weighted)	LwAe	125.0	dB(A)
Sound pressure level of valve (A-weighted)	LpAe	113.0	dB(A)





Comments:

Reliability index - Ri

Sound pressure level: 80 %; High velocity: 20 %

Recommended solution: Heavy duty valve, Hardened seat/plug, Low-noise

Legend

-  Errors
-  Alarm